

Metals and Non-metals

How metals and non-metals look and behave, the reactivity series, how ionic compounds are formed, how metals are extracted from ores, and how to prevent corrosion.

Introduction

We use metals and non-metals every day. Iron and aluminium are used to build houses and vehicles. Copper is used for wires. Sulphur and phosphorus are non-metals used in fertilisers and medicines. In this chapter we compare their properties as in the rationalised NCERT for 2026-27. There are about 118 elements known, and most of them are metals.

Physical Properties

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Physical properties are those that can be seen or measured without changing the substance into another substance.

- **Lustre:** Metals are shiny in pure form. Non-metals are dull, except iodine which is lustrous and graphite which is shiny.
- **Hardness:** Most metals are hard. Sodium and potassium are soft and can be cut with a knife. Non-metals are generally soft, except diamond which is the hardest natural substance.
- **Malleability:** Metals can be beaten into thin sheets. Gold and silver are the most malleable.
- **Ductility:** Metals can be drawn into wires. Gold is the most ductile.
- **Conductivity:** Metals are good conductors of heat and electricity. Silver is the best conductor, followed by copper and aluminium. Lead and mercury are poor

conductors among metals. Non-metals are generally poor conductors, except graphite which conducts electricity.

- **Melting and boiling points:** Metals usually have high melting and boiling points. Gallium and caesium have low melting points and will melt on the palm. Iodine and graphite (non-metals) also have high melting points in some cases.
- **Sonorous:** Metals produce a ringing sound when struck.
- **Density:** Metals generally have high density. Non-metals have low density.

Chemical Properties

1. Reaction with Oxygen (Formation of Oxides)

Metals react with oxygen to form metal oxides, which are basic. Non-metals form oxides which are acidic or neutral.

- $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$ (black) on heating. $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$.
- Metal oxides turn red litmus blue. Dissolved in water they give alkalis: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$, $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$.
- Some metal oxides are amphoteric, meaning they react with both acid and base. Example: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ and $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ (sodium aluminate).
- At ordinary temperature, the surface of aluminium, magnesium and zinc forms a thin oxide layer which protects them from further attack.

2. Reaction with Water

- **Highly reactive:** Sodium and potassium react violently with cold water: $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2 + \text{heat}$. The hydrogen can catch fire. Calcium reacts less violently: $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$.
- **With hot water:** Magnesium reacts with hot water: $\text{Mg} + 2\text{H}_2\text{O}(\text{hot}) \rightarrow \text{Mg(OH)}_2 + \text{H}_2$.
- **With steam:** Aluminium, zinc and iron react with steam: $2\text{Al} + 3\text{H}_2\text{O}(\text{steam}) \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2$, $\text{Zn} + \text{H}_2\text{O}(\text{steam}) \rightarrow \text{ZnO} + \text{H}_2$, $3\text{Fe} + 4\text{H}_2\text{O}(\text{steam}) \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$.
- **No reaction:** Lead, copper, silver and gold do not react with water at all.

3. Reaction with Acids

Metals above hydrogen in the reactivity series react with dilute acids to give salt and hydrogen.

- $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$, $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$.
- With dilute nitric acid, hydrogen is not evolved because HNO_3 is a strong oxidising agent and oxidises H_2 to H_2O . Only manganese and magnesium give H_2 with very dilute HNO_3 .
- Aqua regia is a fresh mix of concentrated HCl and concentrated HNO_3 in the ratio 3:1. It can dissolve gold and platinum.

4. Reaction with Other Metal Salts

A more reactive metal displaces a less reactive metal from its salt solution. This is already seen in Chapter 1. Example: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Worked Example 1, Check Reactivity with Water

Which of these will give hydrogen when added to cold water: Na, Mg, Fe, Cu?

Na reacts violently with cold water to give H_2 . Mg reacts only with hot water, not cold water. Fe reacts only with steam. Cu does not react with water at all. So only Na gives H_2 with cold water.

Reactivity Series

Reactivity Series

The reactivity series is a list of metals in order of decreasing reactivity, decided by how vigorously they react with oxygen, water and acids, and by displacement reactions.

From most reactive to least reactive:



Hydrogen is not a metal but is included for comparison. Metals above hydrogen can displace hydrogen from acids. Metals below hydrogen cannot.

How it is decided: If metal A displaces metal B from its salt solution, then A is more reactive than B. Example: Zn displaces Cu from CuSO_4 , so Zn is above Cu.

Worked Example 2, Use the Series

Predict if these reactions will happen: (a) $\text{Zn} + \text{CuSO}_4 \rightarrow$ (b) $\text{Cu} + \text{ZnSO}_4 \rightarrow$

(a) Zn is above Cu in the series, so Zn will displace Cu. Reaction happens: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$.

(b) Cu is below Zn, so Cu cannot displace Zn. No reaction.

Ionic Compounds

Metals lose electrons and become positive ions (cations). Non-metals gain electrons and become negative ions (anions). The bond formed by transfer of electrons is an ionic bond.

Ionic (Electrovalent) Compound

A compound formed by complete transfer of electrons from a metal to a non-metal, held together by strong electrostatic force between opposite ions.

Formation of Sodium Chloride (NaCl)

- Sodium has electronic configuration 2,8,1. It loses 1 electron to get the stable configuration of neon 2,8. $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$.
- Chlorine has 2,8,7. It gains 1 electron to get the stable configuration of argon 2,8,8. $\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$.
- Na^+ and Cl^- attract each other strongly and form NaCl .

Formation of Magnesium Chloride (MgCl_2)

- Mg is 2,8,2. It loses 2 electrons: $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$.
- Each Cl gains 1 electron, so two Cl atoms are needed: $2\text{Cl} + 2\text{e}^- \rightarrow 2\text{Cl}^-$.
- Mg^{2+} with 2Cl^- forms MgCl_2 .

Properties of Ionic Compounds

- They are solids and hard because of strong force between ions.

- They have high melting and boiling points. Example: NaCl melts at 1074 K.
- They are generally soluble in water and insoluble in solvents like kerosene and petrol.
- They conduct electricity in molten state and in aqueous solution because ions are free to move. They do not conduct in solid state because ions cannot move.

Occurrence and Extraction

Minerals and Ores

Metals occur in the Earth's crust in minerals. A mineral from which a metal can be extracted conveniently and economically is called an ore.

- **Native state:** Very unreactive metals like Au, Ag and Pt are found free in nature.
- **Combined state:** Reactive metals are found as oxides, sulphides or carbonates. Example: Na as NaCl, Mg as MgCO₃, Fe as Fe₂O₃.
- **Gangue:** The sand and rocky impurities present in an ore.

Steps of Extraction (Metallurgy)

Step 1. Enrichment of ore: Remove gangue. This is based on differences in physical or chemical properties.

Step 2. Extraction of metal from concentrated ore: Depends on reactivity.

- **Low reactivity metals (Cu, Hg, Ag):** Their oxides can be reduced by heat alone. Example: $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2$ (roasting), then $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ on heating. $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$, then $\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$.
- **Medium reactivity metals (Zn, Fe, Pb):** Their sulphide or carbonate ores are first converted to oxides by roasting or calcination. Roasting is heating in presence of air for sulphide ores. Calcination is heating in absence of air for carbonate ores. Then oxide is reduced with carbon: $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$, $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$. This is done in a blast furnace for iron.
- **High reactivity metals (K, Na, Ca, Mg, Al):** These cannot be reduced by carbon because they are stronger reducing agents. They are obtained by electrolytic reduction of their molten chlorides or oxides. Example: At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$, at anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$. Overall: $2\text{NaCl}(\text{molten}) \rightarrow 2\text{Na} + \text{Cl}_2$ on electrolysis. Aluminium is obtained from alumina Al₂O₃ by electrolysis.

Step 3. Refining of metal: The metal obtained still has impurities. The most common method is electrolytic refining. Example for copper: anode is impure copper, cathode is pure copper, electrolyte is acidified CuSO_4 . On passing current, pure copper deposits on the cathode and impurities settle as anode mud.

Worked Example 3, Roasting and Calcination

What is the difference between roasting and calcination? Give one equation each.

Roasting is heating a sulphide ore in the presence of air to convert it into oxide. Example: $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$.

Calcination is heating a carbonate ore in the absence or limited supply of air to convert it into oxide. Example: $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$ on heating. Both steps give the oxide which is then reduced to metal.

Corrosion and Prevention

Corrosion was introduced in Chapter 1. For metals, the main example is rusting of iron.

- **Rusting:** Iron reacts with oxygen and moisture: $4\text{Fe} + 3\text{O}_2 + x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (hydrated ferric oxide). Rust is brown and weak. It eats away the metal.
- **Tarnishing:** Silver forms black Ag_2S with H_2S in air. Copper forms green basic copper carbonate $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$.
- **Prevention:**
 - Painting, oiling and greasing keep air and water away.
 - Galvanisation: coating iron or steel with zinc. The coating keeps air and moisture away from the iron and slows corrosion.
 - Alloying: mixing a metal with other metals or non-metals. Example: stainless steel is iron with nickel and chromium. It does not rust.
 - Chrome plating and anodising for aluminium.

Alloys: An alloy is a homogeneous mixture of metals, or a metal with a non-metal.

- Brass: $\text{Cu} + \text{Zn}$. Bronze: $\text{Cu} + \text{Sn}$. Solder: $\text{Pb} + \text{Sn}$ (low melting point, traditionally used for soldering electrical connections).
- Amalgam: alloy that contains mercury.

- 24 carat gold is pure gold. 22 carat gold is 22 parts gold and 2 parts copper or silver, made harder for jewellery.
- Steel: iron with carbon (0.05 to 1.5 percent). Hard and strong.

Summary

- Metals are lustrous, malleable, ductile, sonorous and good conductors. Non-metals are dull, brittle and poor conductors, except graphite and iodine.
- Metal oxides are basic, non-metal oxides are acidic or neutral. Some metal oxides like Al_2O_3 and ZnO are amphoteric.
- Reactivity with water and acids decreases down the series $\text{K} > \text{Na} > \text{Ca} > \text{Mg} > \text{Al} > \text{Zn} > \text{Fe} > \text{Pb} > \text{H} > \text{Cu} > \text{Hg} > \text{Ag} > \text{Au}$.
- Ionic compounds are formed by transfer of electrons from metal to non-metal. They are hard, have high melting points, dissolve in water and conduct electricity when molten or in solution.
- Metals are extracted as per reactivity: low reactivity by heating alone, medium by roasting or calcination followed by reduction with carbon, high by electrolysis of molten ores.
- Refining is often done by electrolytic refining.
- Corrosion is prevented by painting, oiling, galvanisation and alloying.

Practice Questions

1. Name two metals that occur in free state and two that occur only as compounds. Why?

2. What happens when (a) sodium is added to cold water (b) magnesium is added to hot water (c) iron is added to steam? Write equations.

3. Show the formation of MgCl_2 by transfer of electrons. Write the electronic configurations.

4. Why do ionic compounds have high melting points and conduct electricity in molten state but not as solids?

5. Explain why aluminium is obtained by electrolysis and not by reduction with carbon. Give the reaction at the cathode and anode for NaCl.

6. What is galvanisation? Why is zinc used to protect iron and not copper?